Ø=Ý9 2. Entities, Attributes, and KeysÐ
1þ ã TaxPayerÐ
Attribute•@ype”FW67 iptionÐ
taxpayer_id (PK)””åIUnique taxpayer IDÐ
name•`ARCHAR(100)•@axpayer full nameÐ
dob”@ATE”F FR öb &— thÐ
address•`ARCHAR(200)•&W6–FVçF– Â FG&W7=
email•`ARCHAR(100)•Væ— VR VÖ –Â `or loginÐ
phone•`ARCHAR(15)”6öçF 7B àumberÐ
pan_number•`ARCHAR(20)•Væ— VR AN numberÐ
password•`ARCHAR(100)”Æöv–â 70wordÐ
2þ ã IncomeÐ
Attribute•@ype”FW67 iptionÐ
income_id (PK)””åIUnique income record IDÐ
taxpayer_id (FK)””åIRefers to TaxPayerÐ
financial_year•`ARCHAR(10)••ear (e.g., 2023-24)Ð
salary_income”DT4”Ô Âƒ Ã"™Income from salaryÐ
business_income”DT4”Ô Âƒ Ã"™Income from businessÐ
other_income”DT4”Ô Âƒ Ã"™Other sourcesÐ
total_income”DT4”Ô Âƒ Ã"™Total incomeÐ
3þ ã TaxRuleÐ
Attribute•@ype”FW67 iptionÐ
rule_id (PK)””åIUnique tax rule IDÐ
financial_year•`ARCHAR(10)” Æ–6 le yearÐ
min_income”DT4”Ô Âƒ Ã"™Minimum income for slabÐ
max_income”DT4”Ô Âƒ Ã"™Maximum income for slabÐ
tax_rate”DT4”Ô ÂƒRÃ"™Tax percentageÐ
4þ ã AdminÐ
Attribute•@ype”FW67 iptionÐ
admin_id (PK)””åIUnique admin IDÐ
name•`ARCHAR(100)” FÖ–â æ Ö]
email•`ARCHAR(100)” FÖ–â VÖ –Í
password•`ARCHAR(100)” FÖ–â 70wordÐ
5þ ã TaxReturnÐ
Attribute•@ype”FW67 iptionÐ
return_id (PK)””åIUnique return IDÐ
taxpayer_id (FK)””åIRefers to TaxPayerÐ
financial_year•`ARCHAR(10)••ear of filingÐ

filing_date”@ATE•&WGW n filing dateÐ
total_tax_paid”DT4”Ô Âƒ Ã"™Amount paidÐ
refund_amount”DT4”Ô Âƒ Ã"™Refund if overpaidÐ
status•`ARCHAR(30)”f–ÆVB ò Verified / RejectedÐ
rule_id (FK)””åIRefers to TaxRuleÐ
admin_id (FK)””åIVerified by AdminÐ
6þ ã TaxPaymentÐ
Attribute•@ype”FW67 iptionÐ
payment_id (PK)””åIUnique payment IDÐ
return_id (FK)””åIRefers to TaxReturnÐ
payment_date”@ATE• ayment dateÐ
amount”DT4”Ô Âƒ Ã"™Amount paidÐ
mode_of_payment•`ARCHAR(20)”öæÆ–æR ò öffÆ–æ]
transaction_id•`ARCHAR(50)•&V`erence numberÐ
Ø=Ý9 3. Relationships with Cardinality & ParticipationÐ
Relationship Name”VçF—F–W2 –àvolved”FW67 iption”6 &F–æ Æ—G™ParticipationÐ
Files•@axPayer !” TaxReturn”öæR F ‡ yer can file many returns“ N•@axPayer (Mandatory), TaxReturn (Mandatory)Ð
Has•@axPayer !” Income”öæR F ‡ yer can have many income records“ N•@axPayer (Mandatory), Income (Optional)Ð
Makes•@axReturn !” TaxPayment”öæR &WGW n can have multiple payments“ N•@axReturn (Mandatory), TaxPayment (Optional)Ð
Managed_By•@axReturn !” Admin” FÖ–â `erifies many returns”â 1•@axReturn (Optional), Admin (Mandatory)Ð
Based_On•@axReturn !” TaxRule”V 6, &WGW n follows one rule”â 1•@axReturn (Mandatory), TaxRule (Mandatory)